**Module 5**

**Time Series Analysis And Predictive Modeling**

Time Series Analysis refers to a way of data analysis by considering the points after a fixed time period (e.g.; daily, monthly, annual) in order to derive certain pattern, trends to construct forecasts.

Time series analysis and forecasting play the most important role in predicting future trends, behaviours and behaviours using the past data. It assists companies to make informed decision, efficient resource allocation, as well as avoid risks, through its ability to predict market demand, sales volatility, stock changes, and so many others. It also helps to plan, budget and strategize in different fields including finance, economics, health, climatic science, and resource usage and promotes efficiency and competitiveness.

**Properties of Time Series Data:**

Temporal Order: The information is sequenced with time (chronologically).

The observations rely on time (a past observation affects other future observations).

Regular Intervals: The data are taken regularly (i.e., hour-to-hour, day-to-day, month-to-month).

The data points are given as the result of watching or recording something in time, like the price of a stock, the temperature in a room, or the amount of sales being made. Time series data Time series data is typically displayed graphically, with time represented on an axis and the variable of interest on an axis so analysts can determine trends, patterns, and time changes.

The representation of the time series data is commonly performed in the form of a line plot where the time is shown on the horizontal axis (x-axis) and the values of the variable on the vertical axis (y-axis). The graphical representation allows the visual interpretation of trends, patterns and fluctuations in the variable over time to assist in the analysis and interpretation of the data.

## Importance of Time Series Analysis

Forecast Future: Time series analysis can be used to derive the future trend and modeling future behavior of businesses in order to help them to be proactive in their decision making such as anticipating the market need, stock prices and other variables.

Identify Patterns and Anomalies: Time series analysis can also aid in identifying repeat patterns and outliers by scrutinizing series of information points which render insight into underlying dynamics and possible outliers.

Risk Mitigation: Through the identification of risks, the businesses are able to come up with mitigation strategies to deal with risks, and this plays a significant role in the management of risks.

Strategic Planning: Behaviors and patterns derived out of time series help in long term strategic planning, thus making decisions about what to do in the finance world, healthcare and other sectors.

Competitive Advantage: Time series analysis helps companies to better allocate resources both with regard to inventory, staff and monetary resources. Businesses can make decisions based on data and be ahead of the market trends and react to changes to gain a competitive advantage.

**Elements of Time Series Analysis**

**Trend:** Whereas a non-uniform interval does not exist and any deviation of the given set of data is any constant flow of time. The trend would be negative or positive or null trend

**Seasonality:** That is some repetitive or deterministic period change in the data on an ongoing time-line. Would be Bell curve or saw tooth

**Cyclical:** Under which there is no specific time, a lack of certainty in the direction and trend of movement

**Irregularity:** The sudden occurrence/event/scenario and spikes within a brief period of time.

| | Trend | Seasonality | Cyclical | Irregularity |
|---|---|---|---|---|
| Time | Fixed Time Interval | Fixed Time Interval | Not Fixed Time Interval | Not Fixed Time Interval |
| Duration | Long and Short Term | Short Term | Long and Short Term | Regular/Irregular |
| Visualization |  |  |  |  |
| Nature - I | Gradual | Swings between Up or Down | Repeating Up and Down | Errored or High Fluctuation |
| Nature – II | Upward/Down Trend | Pattern repeatable | No fixed period | Short and Not repeatable |
| Prediction Capability | Predictable | Predictable | Challenging | Challenging |
| Market Model |  |  |  | Highly random/Unforeseen Events – along with white noise. |

**Types of Time Series Analysis**

Since time series analysis involves numerous categories or versions of data, there is a situation where an analyst has to construct a complicated model. Nevertheless, not all the variances can be explained by analysts, and they are not able to generalize a particular model to all samples. Too complex models or models that attempt too much are also a fit problem. Failure to fit (lack of fit) or failure to partition random error and true relationship (overfitting) models means that models fail to draw the distinction whereby analysis is skewed and forecasts are inaccurate.

Examples of time series analysis models are:

**Classification:** Investigates and allocates patterns to the data.

**Curve fitting:** Graphs the information in a curve to analyze the relations of variables in the data.

**Descriptive analysis:** Determines the trends, cycles or seasonal variation in the data of time series.

**Explanative analysis:** Tries to grasp the information and internal relations in it, cause and effect.

**Exploratory analysis:** Displays the key features of the time series data, typically in graphic form.

**Forecasting:** It guesses values to come. This is the trend-type which is historically oriented. It applies the past in modeling the future data, forecasting of situations that may occur in future plot points.

**Intervention analysis:** examines how data can be altered by an event.

**Segmentation:** Divides the data into portions to demonstrate the property of source information.

**Data classification**

Time series data can be identified with two broad categories:

The data on **stock times series** implies that the values of the attributes are measured at a defined time, such as a fixed picture of the data as it was during the time.

**Flow time series data** entails a record of the activity of the attributes within a specified duration which is usually a component of the whole and forms a section of the outcomes.

In a time series, the variations may appear randomly in data:

Through **functional analysis**, one will be able to select patterns and relationships in data so as to find significant events.

**Trend analysis** refers to ascertaining uniform change at a particular direction. The two classifications of trends are the deterministic and stochastic trends; the latter is random and cannot be explained, and the former can be traced to its ultimate cause.

**Seasonal variation** refers to occurrences of certain and periodic times in the period of one year. Serial dependence is a situation where subsequent points in time are likely to be linked.

## Time Series Forecasting

Time Series Forecasting refers to a statistical time series method that is able to provide pointer to future series using past series values. Simply put, thinking of it is gauging into the future of data value in time. Time Series Forecasting can be used to Figure out future trends using trends and patterns identified in past records and help in decision-making and planning by leading to educated guesses concerning what is likely to occur.

**Various Time Series Forecasting Algorithms**

**Autoregressive (AR) Model**: Autoregressive (AR) model is a model included in the time series with the help of which further values can be predicted on the basis of linear combinations of prior values of the same time series. AR(p) A time series is modeled in an AR(p) model as a linear combination of its past p values, plus a random error term. Autoregressive model Because it is the order of the models (p), there is the number of previous values that are included in the prediction process.

**Autoregressive Integrated Moving Average (ARIMA**): ARIMA models are popular models of time series forecasting. It estimates the following term in a time series with the combination of its past values as well as the previous forecast errors. The parameters of the model are order of autoregression (p), differencing (d) and moving average (q).

**ARIMAX:** extension of ARIMA to incorporate exogenous variables that may result in a more accurate set of forecasts.

**Seasonal Autoregressive Integrated Moving Average (SARIMA):** SARIMA is an extension of ARIMA where seasonality is accounted in the model. It has extra seasonal parameters (P, D, Q) to repay periodic changes in the data.

**SARIMAX:** An augmentation of SARIMA used with seasonal time series forecasts that uses exogenous variables.

**Vector Autoregression (VAR) Models:** VAR models are autoregression of multivariate time series data where each of the variables is a linear combination of its own past values and the past values of the other variables. They become applicable when there is a need to examine and predict interdependencies between several time series.

**Theta Method:** The most basic and the obvious forecasting method that is obtained through extrapolation and trend approximation.

**Exponential Smoothing Techniques:** Exponential smoothing techniques, according to the names Simple Exponential Smoothing (SES) and Holt-Winters, predict the upcoming values with exponentially declining weighting on historical values. Such approaches are especially helpful when dealing with trended and seasonal data.

**The Gaussian Processes Regression:** Gaussian Processes Regression is another Bayesian non-parametric method of modeling functions distributions through time. It gives the uncertainty estimates together with point forecasts, hence it is applicable in capturing the time series forecast uncertainty.

**Generalized Additive Models (GAM):** is a general modeling tool that uses additive components, which makes it possible to have a nonlinear relationship and interactions.

**Random Forests:** Random Forests is a method of related machine learning where multiple decision trees are grown during training and the average of the predictions of the multiple trees are returned. It is suitable to forecast using time series as the data can contain complicated relationships and interaction between the data.

**Gradient Boosting Machines (GBM):** GBM represents the next ensemble-based learning method which by sequentially constructing several decision trees, where each one can rectify the more recent mistakes of the preceding tree. It has excellent performance and has high resistance to overfitting in determining nonlinear relationships.

**State space models:** These state space models are a description of a time series as a mixture between unobserved (hidden) states and observed measurements. These models reflect in

reference to the deterministic and stochastic part of the time series, which means that they can be used in forecasting and observation of any anomalies.

**Dynamic Linear Models (DLMs):** DLMs are a state-space Bayesian model that use a model of time series data which is a representation of a latent state. They are flexible in that they can absorb different trends, seasonality and other moving trends in the data.

**Recurrent Neural Network (RNNs) and Long Short-Term Memory (LSTM) Networks:** RNNs and LSTMs allow working with sequential data, which is another deep learning structure. They are able to learn intricate time-varying patterns in time series data and hence are effective in forecasting problems particularly on large scale and high dimensional data.

**Hidden Markov Model (HMM):** An Hidden Markov Model (HMM) is a mathematical model that describes the sequential results of some observable events as a combination of some hidden states. HMMs enable the inference of unobservable states with respect to observed data within time series and allow defining the dependency and state changes between states. They find useful application in such activities as speech recognition tasks, gesture analysis, and anomaly detection, where they present a framework to learn complex sequential data and discover patterns within that data, leading to meaningful constructions of the data.

## Importing and Cleaning Time Series Data

**Importing:**

Read CSV, Excel, databases or APIs using libraries such as Pandas.

```
import pandas as pd

df = pd.read_csv('timeseries.csv', parse_dates=['Date'], index_col='Date')
```

**Cleaning:**

Treat missing items, duplications, wrong timestamps.

Make datetime an index of the DataFrame.

submit data into proper frequencies (e.g. daily, monthly) with .asfreq() or resample().

### Plotting

The graphical representation of data that has been collected over time intervals forming a chain is the time series visualization. It includes a wide range of such methods as line plots, seasonal subseries plots, autocorrelation plots, histograms and interactive visualization. The techniques are useful in enabling the analysts to detect trends, patterns, and anomalies in time-varying data in order to understand and make better decisions.

Various graphs of visualization of Time series

Line Plots: The line plots are plotted based on a trend, a cycle, and changes in data patterns, which can be easily observed.

Seasonal Plots: Seasonal plots partition time series data into seasonal and non-seasonal data; this enables one to examine trends over a particular period of time.

Histograms and Density Plots: Plots the values of the data and the time of occurrence, thus giving us ideas about the nature of data such as skewness and kurtosis.

Autocorrelation and Partial Autocorrelation plots: Plots to help discover seasonality and lagged relationships as well as time series autocorrelation.

Spectral Analysis Spectral analysis tools, like periodograms and spectrograms, represent the frequency components of a time series data and can be valuable to detect periodicity and cyclical trends.

Decomposition Plots: Decomposition plots separate a time series into its trend, seasonal, and residual prices in order to determine patterns underlying a data set.

Use **Matplotlib** or **Seaborn** for line plots.

```
import matplotlib.pyplot as plt
df['Value'].plot(title='Time Series Plot')
plt.show()
```

## Moving averages

Traders resort to different indicators to assist in making buy and sell decisions in technical analysis. The moving average is one of such indicators that can be utilized to smooth movements in prices on a daily basis to aid in developing a trend on the daily pricing by developing an average price that remains continuously updated.

The effects of short-term random fluctuations in the price of a stock on a given period of time can be reduced by computing the moving average. Simple moving averages (SMAs) incorporate the use of a simple arithmetic average of prices over this period of time whereas the exponential smoothing (EMAs) applies heavier weightings on more recent prices than old prices over the timespan.

Calculation of Moving averages is done to determine the direction of a trend of a stock or to prefer its support and resistance values. It is also trend or lagging indicator since it is pegged on the historical prices.

- Irregularities the short returns, and emphasizes the tendencies.

• Types:

**Simple Moving Average (SMA)**:

A simple moving average (SMA) is determined by the arithmetic mean of a certain the set of values over a period of time. A tranche of figures or prices of stocks are added and the sum is divided by the number of prices in the set. A security has an average security value that can be calculated using the following formula:

```
df['SMA_3'] = df['Value'].rolling(window=3).mean()
```

$$SMA = \frac{A_1 + A_2 + \ldots + A_n}{n}$$

**where:**

$A =$ Average in period $n$

$n =$ Number of time periods

**Exponential Moving Average (EMA)**:

The exponential moving average puts larger weights on recent prices as this is supposed to make them adapt more easily to new information. In order to compute an EMA the simple moving average (SMA) over certain time period is computed.

Then take the multiplier coefficient which scales the EMA, or so-called the smoothing factor, which is usually, [2/(chosen time period + 1)].

```
df['EMA_3'] = df['Value'].ewm(span=3).mean()
```

$$EMA_t = \left[V_t \times \left(\frac{s}{1+d}\right)\right] + EMA_y \times \left[1 - \left(\frac{s}{1+d}\right)\right]$$

**where:**

$EMA_t =$ EMA today

$V_t =$ Value today

$EMA_y =$ EMA yesterday

$s =$ Smoothing

$d =$ Number of days

**Missing values**

Missing values or the data that is not (or not being) stored (or not being present) of some variable/s in the given dataset is called missing data.

Value that is missing in a data may be recorded in many forms, depending on the origin of the data and the conventions. Some of the typical representations here are:

NaN (Not a Number): Missing data is expressed as NaN in a lot of programming languages and data analysis packages. That is the default library of libraries such as Pandas in Python.

None or NULL: Different programming languages and databases usually have missing values, which are marked as None or NULL. An example is SQL databases where a missing value is supposed to be noted using NULL.

The values or data that are not stored (or not there) to some variable/s in the given dataset are called the missing data.

In some databases missing values may be marked by a special symbol, in others a given code can be used which reflects that there is no current value. These are some typical representations:

There might be various explanations in the absence of validations of particular values in the data. The cause of data missing on the dataset influences the process of missing data processing. Therefore, there is the need to learn what might be the reason why the data might be absent.

Among some of them are the reasons below:

Information stored in the past may be affected by failure to maintain.

There are no observations done in some of the fields because of some reasons. This may lead to inaccuracy of recording of the values since it works through humans.

Intentional failure of the user in providing the values is not provided

Item nonresponse: It is an indication that the participant did not participate.

Missing Values: In some cases, the symbols used to represent a missing value include empty strings (""). This usually happens in the data that is given in text format or CSV where a field may be empty.

Special Indicators: Datasets may consider these specific values to represent special indicators, such as -999, 9999, or other values that are unlikely to occur, indicating that it is a special value, i.e. that the data point is missing. Many times it can be observed with older datasets or in certain industries where such conventions have been provided.

Blanks or Spaces: Missing values could also be indicated by spaces or blank fields in some instances especially in the text files whose widths are fixed**.**

- Strategies:
    - **Forward fill**: `df.fillna(method='ffill')`
    - **Backward fill**: `df.fillna(method='bfill')`
    - **Interpolation**: `df.interpolate(method='linear')`
    - **Dropping rows**: only if missing data is minimal
- **Visualize** missing data using `missingno` or `seaborn.heatmap`.

## Serial correlation

Serial correlation Within a time series, serial correlation is the property of a variable and a lagged version of the same variable (i.e. a variable at time T and a variable at time T-1) being found to be correlated with each other over time. Serial correlation is common in repeating patterns when the level of one variable influences future level of the variable. The technical

analysts employ this correlation in financial terms to know how effectively the values of the prices of a security in the past are able to predict the future of the price of the same security.

Serial correlation is analogous to the statistic lagged or auto correlation.

Serial correlation can be defined as a statistical term used to explain the correlation that exists amidst the observations of identical variable regarding particular periods. When the serial correlation of a variable is found at zero then there is no correlation, and all the observations are independent of the other. On the other hand, when the serial correlation of one variable is biased towards 1, then the observations are serially correlated and prior observations influence the outcome of the future observations. In essence, serial correlation variable is not random but has a pattern.

Serial correlation as applied to engineering has the purpose of measuring how a signal, (eg, a signal sent out by a computer or a wave in the radio band) varies relative to itself as a function of time. This idea increased its popularity in economic circles where economist and econometricians used the measure to compare economic data over periods of time.

Nearly every major financial organization currently employees quantitative analysts, quants, to aid in mathematical analysis of some aspect of their business. Technical analysis and other statistical inferences are what these financial trading analysts apply in analyzing and forecasting on the stock market. With the aim to make better predictions and opportunities of a strategy being successful in terms of profitability these modelers are trying to determine the form of the correlations. Also, determining correlation structure more accurately enhances the reality of any of the simulated time series to be based on the model. Precise simulations decrease the danger of the investments strategy.

The success of many of these financial institutions depends on quants because they offer the institution market models which in turn constitutes a foundation of investment affair of the institution.

## Autocorrelation

Autocorrelation A mathematical description of the extent to which a particular time series resembles a lagged realization of the same series with respect to successive time span. It is analogous to the correlation between two different time series but the two time series used in autocorrelation are the same but one series is lagged one or more time periods.

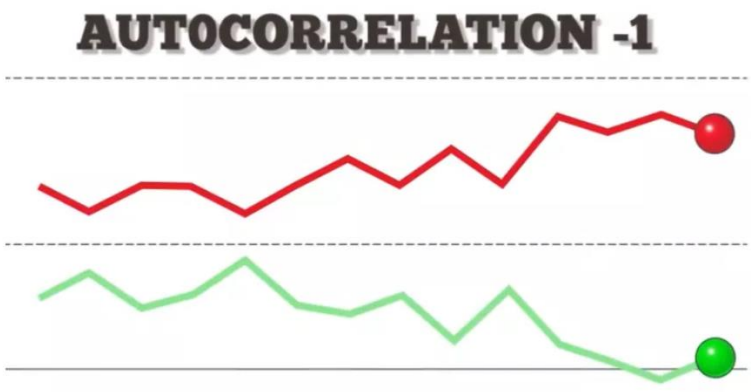
To give an example, assuming that it is raining today, the data will indicate that there is a higher probability that it will rain tomorrow as compared to today when it is clear. In the case of investments, equity is the stock that may exhibit powerful positive autocorrelation of returns to indicate that when the stock is up today, it is much more probable that it would be up tomorrow as well.

Of course, autocorrelation may be an effective concept that can be applied by traders; especially by technical analysts.

Autocorrelation is also called lagged correlation and serial correlation since it evaluates how a current value of a variable relates to its preceding values. Just to give a very simple illustration and look at the five values in the chart below in percentages. These we compare them with the column on the right, which has identical set of values, simply shifted up one row.

| Day | % Gain or Loss | Next Day's % Gain or Loss |
|---|---|---|
| Monday | 10% | 5% |
| Tuesday | 5% | -2% |
| Wednesday | -2% | -8% |
| Thursday | -8% | -5% |
| Friday | -5% | |




**Predictive modeling – Overview**

Predictive modeling is a mathematical process that is employed in the prediction of the future by identifying the trends in a particular and available input data. It forms part of a very important element of predictive analytics, a form of data analytics that is used to predict activity, behavior and trends by using both current and historical data.

Computing the quality of a sales lead, the probability of spam or the probability a person will click on a link or purchase a product are examples of a predictive models. The functionality is frequently embedded within various business applications thus it may be worth getting to know the mechanics of predictive modeling operations in order to do troubleshooting and optimize its functionality.

Despite the implication of the term predictive modeling (to be concerned with the prediction of the future), it can be used to make predictions (e.g., the likelihood that a transaction is fraudulent). Here the act of fraud has already taken place. The idea, here is to forecast that the transaction will be considered as fraud in the future analysis. The predictive modeling is also able to predict future requirements or what-if analysis.

## Evaluating Predictive models

The current data-driven solution engines are the machine learning models, and the question is how would one know whether they are performing their duties, i.e. working well? The model evaluation metrics would play this role: assisting a person in grasping the strengths and weaknesses of a model with the aim of optimization and final implementation of the matter in real life. Whether it is a model to store which emails should be classified, house price forecast, customer segment clustering, among others, learning how to assess a model based on different metrics is a valuable job.

A confusion matrix is a visualization of prediction performance by categorizing outcomes:

- **True Positive (TP)**: Correctly identified as positive
- **True Negative (TN)**: Correctly identified as negative
- **False Positive (FP)**: Incorrectly identified as positive
- **False Negative (FN)**: Missed positive cases

| # | Predicted Positive | Predicted Negative |
|---|---|---|
| **Actual Positive** | True Positive (TP) | False Negative (FN) |
| **Actual Negative** | False Positive (FP) | True Negative (TN) |

Accuracy, defined as (TP+TN)/(Total), is simple. However it can be misleading in an unbalanced dataset.

$$\text{ACCURACY} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision: Also known as Positive Predictive Value, it is the value that shows the true positive cases calculated against a total number of predicted positive outcomes.

$$\text{PRECISION} = \frac{TP}{TP + FP}$$

Recall (Sensitivity or True Positive Rate): It measures the percentage of the true positive out of all the real positives.

$$\text{RECALL} = \frac{TP}{TP + FN}$$

F1 Score: It measures a range between precision and recall, and gives an equal balance to the trade-off between precision and recall.

$$F1 = \frac{2 \times \text{PRECISION} \times \text{RECALL}}{\text{PRECISION} + \text{RECALL}}$$

ROC-AUC (Receiver Operating Characteristic -Area Under Curve)

The ROC curve plots the relationship between False Positive Rate (FPR) and True Positive Rate (TPR) at each of different threshold levels of classification. The AUC (Area Under Curve) condenses the capacity of the model to differentiate between the classes:

AUC 1.0: The ideal classifier

ABC

AUC= 0.5: Random accuracy

Regression Metrics: In predicting continuous values

Root Means Error squared (RMSE): Standard deviation of residuals:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{j=1}^{n}(y_j - \hat{y}_j)^2}$$

Mean Absolute Error (MAE): Simplicity at Work

MAE is the average of the absolute differences between the predicted and actual value has been calculated.

$$\text{MEAN ABSOLUTE ERROR} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|$$

Mean Squared Error (MSE)

$$\text{MEAN SQUARED ERROR} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|^2$$

As RMSE without the square root:

$$\text{ADJUSTED R}^2 = 1 - \frac{(1 - R^2)(N - 1)}{N - p - 1}$$

## Building Predictive Model solutions

Predictive analytics has received the status of a separate area within the scope of data analysis, which employs a range of tools, including statistics, machine learning, and artificial neural networks towards a possible forecasting of the future course of events, relying solely on the past data.

The primary objective of predictive analytics is expressing what could occur in the future, such as trends, particular behavior and events, amongst others, to automatically classify data points, or anticipate possible outcomes with data. In simple terms, predictive analytics involves utilizing all the information in determining the probability of occurrence.

In addition to the above, there are other forms of analytics that you have probably heard of, i.e. descriptive analytics, diagnostic analytics, and prescriptive analytics, which are simply the levels of analytics maturity.

However, before immersing yourself into data and algorithms, it is crucial first to determine the kind of issues you wish to overcome through the predictive analytics model. This could be through a consultation with stakeholders who you are looking to hear about their frustrations, and what their expectation and success would be like.

**Data collection**

After having defined the goals, you continue to gather information to generate a model of a predictive analysis. In the case of data, most of the time it falls into 2 categories:

Scenario 1 -Your company already has sufficient internal business data that has been accumulated over the years.

Scenario 2 - There is no data at your disposal or the amount of it is not copious. In case it is so, it will be required to include data based on external sources (e.g. free public datasets; some data can be obtained on providers).

**Processing and preparation of data**

The most crucial and lengthiest stage of the AI predictive modeling segment which can be described as data preparation and processing. The stage may consume as much as 80% of the total project time since detailed work is involved.

**Data cleaning**

The step to take is to eliminate mistake and discrepancies. This is in the sense that we:

remove noise

correct inaccuracies

Data that are blanks can be replaced by average numbers

delete un finished or abnormal records, etc.

Clean data will help to eliminate low-quality inputs that may distort predictions of the model.

**Data transformation**

Once cleaned the data must often be converted into an appropriate model-ready format. In this step we approach things in the following way.

Feature engineering

It is the method of applying domain knowledge to filter, transform or construct new characteristics out of raw data so as to aid the predictive capacity of machine learning algorithms. In essence, all we evaluate is the usefulness of the current available data features and concur to determine whether new data features are required or not.

This will need that a deep analysis of the data is carried out to find the latent correlations and it will identify the importance of every feature.

To illustrate, we may not use height and weight as two different features when manipulating the data of patients, but we form a new variable, known as Body Mass Index (BMI).

Training and test set generation

The last procedure is to split the information into distinct groups: traditionally 80 percent will be used to train and 20 percent to test and verify. Such division is used to train the model correctly and measure its performance properly.

Choicing of predictive modelling strategies

When we possess some data we choose some predictive modeling method. Different predictive models can be selected, depending on a task you want to solve. And, as the example it is possible to put before us Scikit-learn library where even in the separate roadmap the idea of what model better to choose according to what and how much data is already at hand and what objective is can be grasped.

Classification

The classification of machine learning is merely the forecast of group or category of the data under consideration. It receives input data, and a category (class) label is applied to these data depending on the features they have.

nce we select the appropriate future model, we train it using your ready dataset. During this phase, you set a specific parameter - the configuratory elements that determine how the model behaves. For example, a healthcare model that predicts the risks of lung cancer can control the rate of learning or model to prevent overfiting.

We also need to adjust these parameters to improve the accuracy and efficiency of the model. For example, in e-commerce applications for predicting demand, these adjustments may be favorable for handling seasonal trends or changes in consumer behavior.

In addition, when we train a future stating model, we assess its performance against testing data using matrix that are suitable for its type.

For classification tasks, we use matrix such as:

The accuracy assesses the overall purity of the model.

The accurate checks how many examples of the model predict correctly.

In regression functions, normal matrix includes:

The mean paid error (MSE) calculates the average repayment difference between estimated values and real values.

R-Squar shows how well the model predicts unseen samples.

Of course, there are many other machine learning matrix to include.

## Sentiment Analysis

The emotion analysis, also known as opinion mining, involves subjective information such as processing text data to remove emotions, opinions or approaches. This enables machines to explain the spirit of the text material by analyzing keywords, references, tones and grammar.

For example, analyzing product reviews to determine that customers are satisfied or dissatisfied, it is a general use of emotion analysis.

**Importance**

Customer insights: Businesses help businesses understand the opinion of customers about products or services.

Brand Monitoring: Tracks public spirit about a brand on platforms such as social media.

Market Research: Analyze the trend and public opinion to inform the decision making.

Crisis Management: Recognizes negative emotions quickly, allows organizations to address issues immediately.

Individualization: Customer enables corresponding marketing campaigns depending on emotions and preferences.